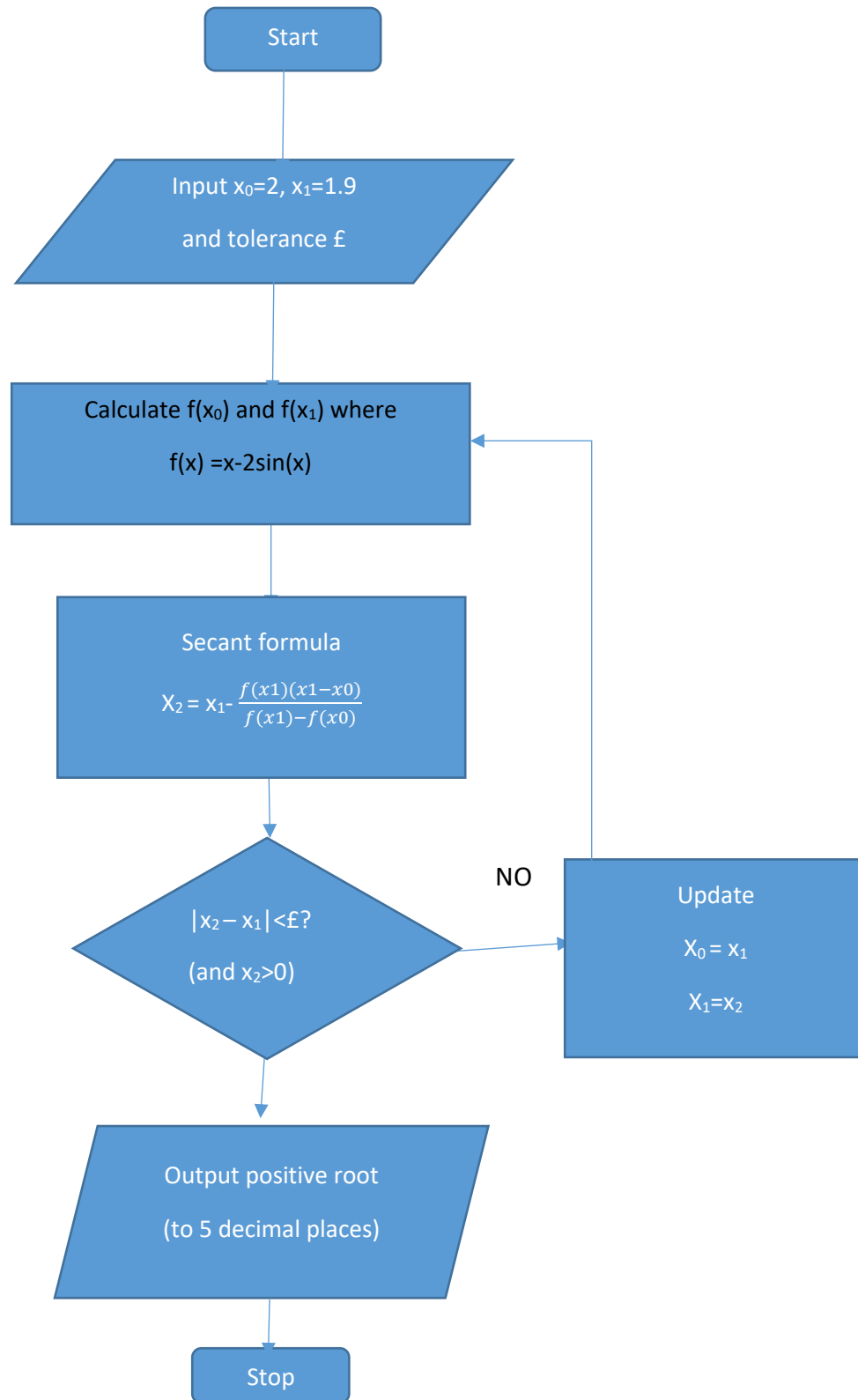


## SECANT METHOD

Find the positive solution of  $f(x)=x-2\sin(x)=0$  by the secant method, starting with  $x_0=2$  and  $x_1=1.9$



## **PSEUDOCODE**

### **START**

Define  $f(x)=x-2\sin(x)$

Set  $x_0=2$

Set  $x_1=1.9$

Set tolerance  $\epsilon$

### **REPEAT**

    Calculate  $f(x_0)$

    Calculate  $f(x_1)$

$x_2=x_1-f(x_1)*(x_1-x_0)/(f(x_1)-f(x_0))$

    IF  $|x_2-x_1| < \epsilon$  AND  $x_2 > 0$

**STOP**

ELSE

$x_0=x_1$

$x_1=x_2$

END IF

UNTIL convergence

Display positive root

**STOP**

## MATLAB CODE

```
f=@(x)x-2*sinx(x);
x0=2;
x1=1.9;
tol=1e-6;
maxiter=100;
for i=1:maxiter
    x2=x1-f(x1)*(x1-x0)/(f(x1)-f(x0));
    if abs(x2-x1)<tol
        break;
    end
    x0=x1;
    x1=x2;
end
```

## ‘FOR’AND ‘WHILE’ LOOPS

```
fprintf('positive root=%.5f\n',x2);
fprintf('Number of iterations =%d/n',i);
```

```
f=@(x)x-2*sin(x);
x0=2;
x1=1.9;
tol=0.000001;
N=100;
for i=1:n
    x2=x1 -f(x1)*(x1-x0)/(f(x1)-f(x0));
    if abs(x2-x1)<tol
        break
    end
    x0=x1;
    x1=x2;
end
fprintf('positive root=%.5f\n',x2);
fprintf('Iterations =%d/n',i);
```

```
while i<N
    X2 =X1 -f(x1)*(x1-x0)/(f(x1)-f(x0));
    i=i+1;
    if abs(x2-x1)<tol
        break
    end
    x0=x1;
```

```
        x1=x2;  
end  
fprintf('positive root =%.5f/n',x2);  
fprintf('Iterations= %d/n',i);
```